

Joshua Negreanu

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PhD student building real-time robot learning systems and generative models, with production experience in teleoperation infrastructure, diffusion models, and large-scale training on DGX systems.

EDUCATION

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- Oregon State University** Corvallis, Oregon
 - PhD Artificial Intelligence (*Accelerated*), GPA: 4.00 Sep 2024 - Jun 2028
Relevant Coursework: Deep Learning, NLP with Deep Learning, ML, Online ML, Optimization, Algorithms and Data Structures.
 - Oregon State University** Corvallis, Oregon
 - BS Computer Science (*AI focus*), GPA: 4.00 Sep 2021 - Jun 2025
 - BS Mathematics (*Applied focus*), GPA: 4.00 Sep 2021 - Jun 2025
Relevant Coursework: OS I and II, ML, Parallel Programming, Applied PDEs, Systems of ODEs, Computational Number Theory.

SKILLS SUMMARY

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- Languages:** Python, C/C++, JavaScript, Bash, SQL, Java, Elm
 - ML/DL:** PyTorch, Hugging Face (LeRobot, Transformers), NumPy, SciPy, Scikit-learn, Wandb
 - Systems/Acceleration:** CUDA, OpenMP, OpenCL, GStreamer, Linux, Git, Docker
 - Data/Infrastructure:** Pandas, Matplotlib, OpenCV, AWS, Azure, DGX, Jetson
 - Backend:** Node.js, mediasoup, WebXR

EXPERIENCE

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- Revobots** Remote
 - AI Intern* June 2025 - Feb 2026
 - Remote Teleoperation:** Designed and deployed mediasoup SFU for low-latency (30 fps) robot vision and control streaming, supporting 500+ concurrent clients per robot.
 - VR Embodiment:** Implemented 60-fps 4K stereo streaming with WebXR headset control and GStreamer GPU encoding for real-time VR embodiment and dataset collection.
 - Data Collection:** Built end-to-end data pipeline using Hugging Face LeRobot for teleoperation recording, collecting 100+ episode datasets with automated compression, AWS storage, and metadata tagging.
 - Model Training:** Designed MLOps pipeline for automated multi-day training of transformer-based robot manipulation policies on DGX system, achieving reliable task execution from collected teleoperation datasets.
 - Oregon State University** Onsite
 - Educational Assistant* Sep 2022 - Feb 2026
 - Teaching:** GTA/ULA across four courses (Programming Language Fundamentals, Operating Systems, Data Structures, General Engineering), serving 100+ students total with office hours, grading hours, and recitations.
 - Tooling:** Implemented autograder scripts for unit testing functional programming assignments in Haskell and Prolog, reducing manual grading time.

RESEARCH

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- Graduate Thesis (In Progress):** Investigating reinforcement learning methods for reasoning and agentic decision-making in discrete diffusion language models, with applications to multi-step planning. Under the mentorship of Huazheng Wang.
 - Undergraduate Thesis: Attacks on Stochastic Dueling Bandits:** Proposed a mathematical framework for reward poisoning algorithms in the stochastic dueling bandits setting, demonstrating vulnerabilities that motivated the development of corruption-robust algorithms. Under the mentorship of Huazheng Wang. Paper available on *OSU Archives*.

PROJECTS

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- Thompson Sampling with Diffusion and Flow Matching Priors for Linear Bandits:** Developed guided diffusion and flow matching algorithms for linear bandit prior sampling, as well as low-rank approximations to reduce per-round time complexity while preserving regret bounds. Open-source code available on GitHub: <https://github.com/joshnegreanu/diffusion-priors>
 - Discrete Diffusion and Auto-Regressive Language Model:** Built both auto-regressive and discrete diffusion transformer language models from scratch. Compared generation quality and inference speed across architectures on openwebtext using DGX training. Open-source code available on GitHub: <https://github.com/joshnegreanu/language-diffusion>
 - Denoising Diffusion and Flow Matching Image Generation:** Implemented denoising diffusion and flow matching generative models (CNN U-Net and latent vision transformer) from scratch. Trained on Stanford Cars, CelebA, and ImageNet on DGX system. Open-source code available on GitHub: <https://github.com/joshnegreanu/image-diffusion>

HONORS AND AWARDS

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- CV Copson Scholarship:** Awarded at the annual OSU Lonseth Lecture, May 2025
 - WIC Culture of Writing Award:** Awarded at the annual OSU Lonseth Lecture, May 2025
 - Dean's List:** Sep 2021 - June 2025